Transponder Landing System (TLS) – System Overview

This document summarizes how the TLS works, names the antenna units, describes their roles, and illustrates the end-to-end signal path (receive → process → transmit). Images are extracted from the provided ANPC manuals for technical fidelity.

Antenna Units – Quick Reference

Unit	Full Name	Role	Signal Direction
ASA	Azimuth Sensor Assembly	Measures horizontal angle / AOA & TOA for azimuth	Receive (1090 MHz)
ESA	Elevation Sensor Assembly	Measures vertical angle / AOA & TOA for elevation	Receive (1090 MHz)
ATA	Azimuth Time-of-Arrival	Aux antenna aiding TOA in azimuth plane	Receive (1090 MHz)
Cal/BIT	Calibration / Built-In-Test Transmitter	Injects known test pulses for calibration and integrity	Transmit (1090 MHz test)
Uplink	Localizer & Glide Slope Transmit Antennas	Broadcast ILS-like guidance to aircraft	Transmit (VHF LOC / UHF GS)
Interrogation	Interrogation & (optional) Suppression	Interrogates Mode A/C transponders; shapes service volume	Transmit (1030 MHz); Receive (1090 MHz)

How TLS Works (High Level)

TLS interrogates aircraft transponders at 1030 MHz; aircraft reply at 1090 MHz. Receiver arrays (ASA/ESA and ATA) measure Angle-of-Arrival (AOA) and Time-of-Arrival (TOA) to estimate aircraft position in 3D. The base station computes ILS-equivalent guidance (localizer and glide slope) targeted to the cleared aircraft and transmits it via the uplink antennas. The guidance appears to the aircraft as a standard ILS (ICAO Annex 10 Cat I compliant).

End-to-End Signal Path

1) Interrogate (1030 MHz) → 2) Aircraft replies (1090 MHz) → 3) Sensors measure AOA/TOA → 4) Base computes guidance → 5) Uplink transmits ILS-like LOC/GS to the aircraft.

Cal/BIT	Calibration/Built-in-Test Transmitter
RCU	Remote Control Unit for entering transponder codes and viewing system status
Uplink	VHF and UHF guidance signal antennas
Interrogator	Interrogation antenna and optional suppression antenna

2.2.1 Maintenance Interface (MI)

The Maintenance Interface (MI) [Figure 2-1] is a program which runs in the Windows operating system. It is used to start the system software, configure and calibrate the system, as well as allow a technician to check the system's health and status and facilitate trouble shooting.

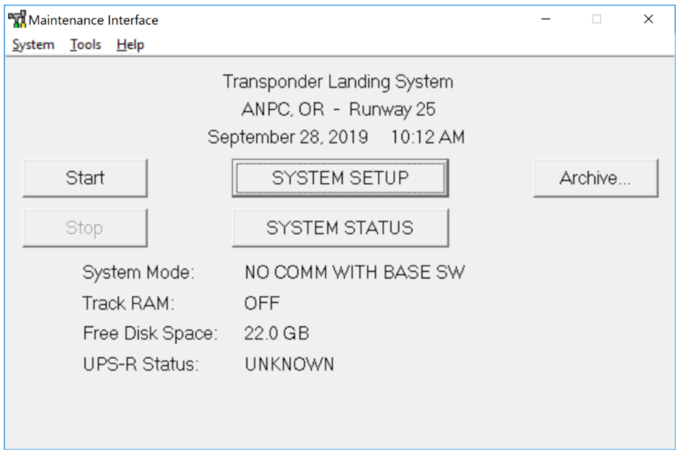


Figure 2-1 Maintenance Interface (MI) Window

2.3 TLS Coverage Volumes

The TLS has two coverage volumes that define how it operates.

1. The surveillance coverage volume (SCV) within which the system tracks aircraft that are within line-of-sight and reply to TLS interrogations. The surveillance track data is displayed on the RCU. Aircraft beyond the SCV are not detected by the TLS. The coverage volume extends to between 60 and 120 miles depending on the version of the TLS sensors installed and type of interrogation antenna.
2. The guidance coverage volume where localizer and glide slope guidance signals can be received by the aircraft. The TLS guidance coverage volume complies with the ICAO standard for ILS. The guidance coverage volume meets and exceeds the standard coverage volumes required by ICAO and extends in some cases much further to meet the requirements for signal coverage for the Instrument Approach Procedure (IAP).

Signal flow / software environment excerpt (Operators Manual p.11)

ASA – Azimuth Sensor Assembly

Measures horizontal AOA and TOA to determine lateral (azimuth) position. Mounted laterally offset from runway centerline. Feeds the base for course computation.

Table 2-1 Summary of key siting parameters

	Minimum	Standard	Maximum
ASA w.r.t Touch Down (TD)	TD-100m	TD-50m	TD+75m
ASA to ESA separation	50 m (option 30 m)	100 m	No spec
ESA to Cal/BIT separation	150 m	150 m	No spec
ASA to ATA	50 m	100	150 m
ESA to ATA	50 m	100	150 m
ATA to Cal/BIT	20 m	60 m	No spec
Interrogator to shelter	11 m	12 m	13m
ASA to ESA angle from perpendicular to centerline	0 degrees	5 degrees	10 degrees
ASA to centerline	50m or as OFZ permits	60 m	70m
ESA to centerline	As permitted by local regs	90 m	200 m

For any parameter noted as “no spec” the practical considerations of property boundaries and cable lengths will impose a maximum limit.

2.2 Examples of Common Siting for TLS Components

Figure 2-3 All TLS equipment on one side of runway

ASA – Azimuth Sensor Assembly (Installation Manual p.11)

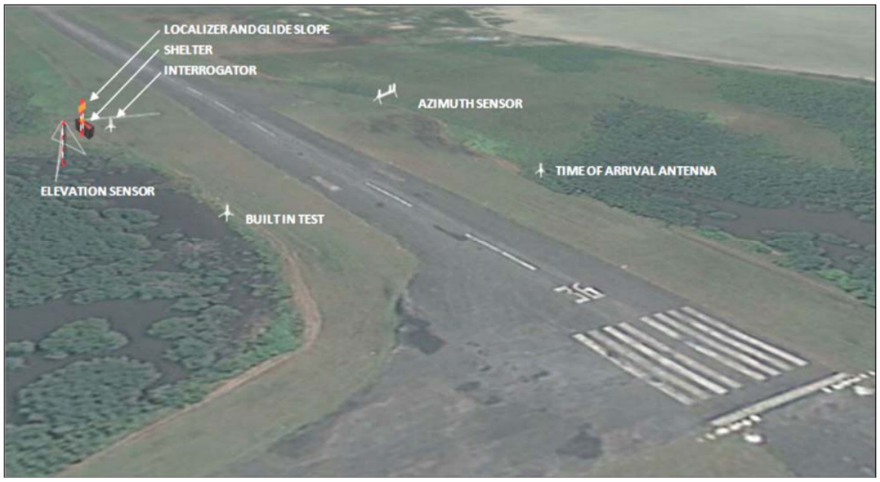


Figure 2-4 TLS equipment split on both sides of the runway

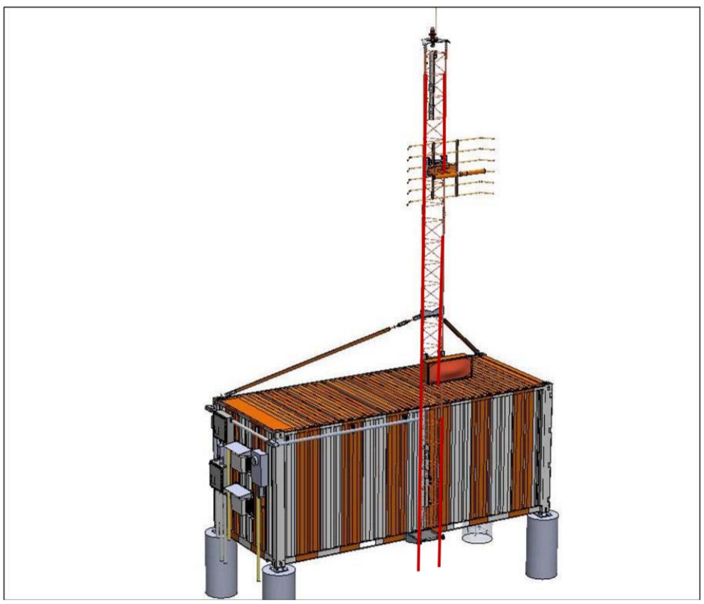
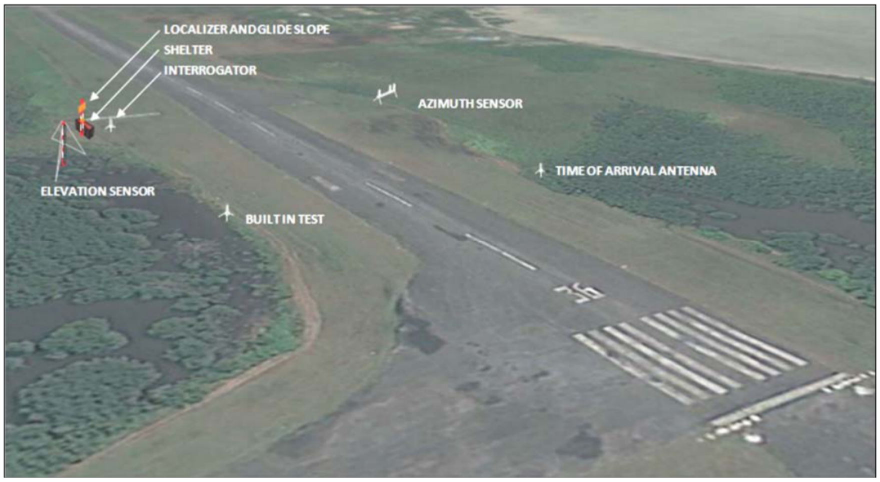


Figure 2-5 TLS shelter with hinged uplink antenna

ASA – Azimuth Sensor Assembly (Installation Manual p.12)

ESA – Elevation Sensor Assembly

Measures vertical AOA and TOA to determine elevation. Mounted on a tower; defines critical/sensitive areas due to height.



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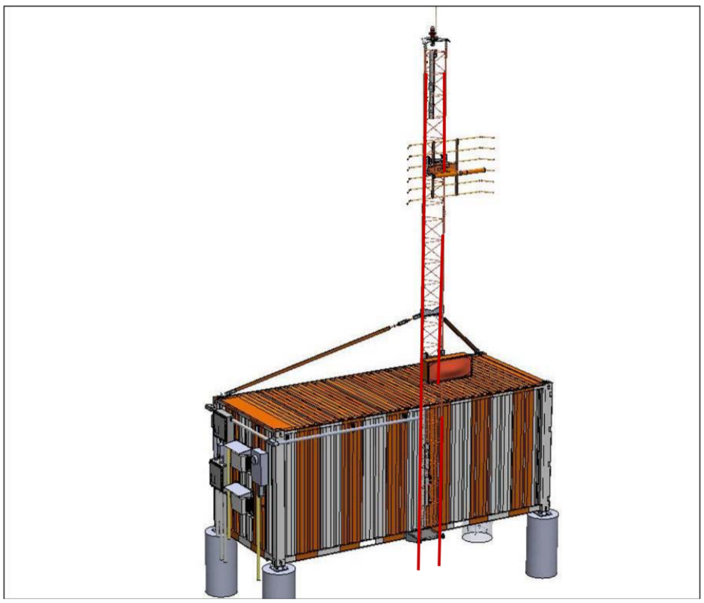

Figure 2-5 TLS shelter with hinged uplink antenna

ESA – Elevation Sensor Assembly (Installation Manual p.12)

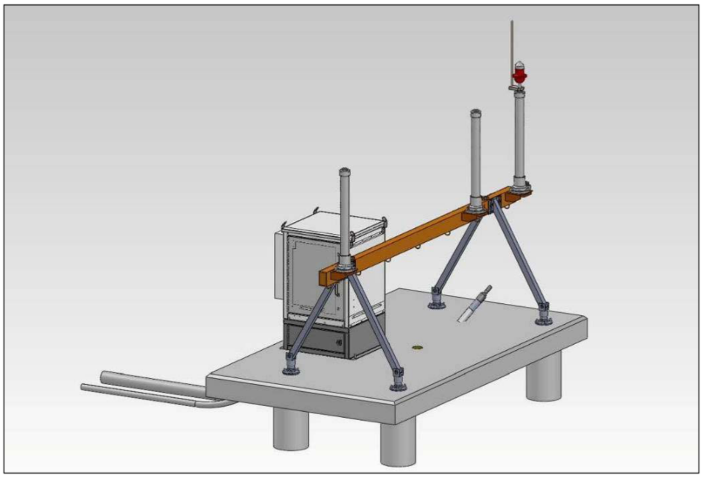

Figure 2-6 ASA with omni directional antennas

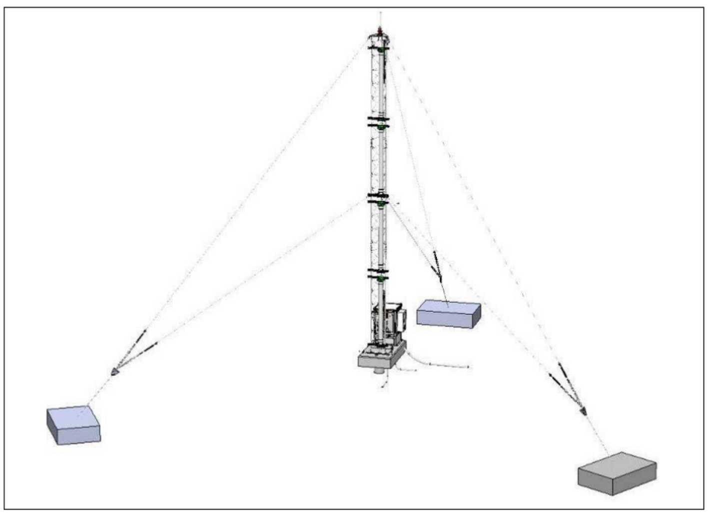

Figure 2-7 ESA with guy wire option

ESA – Elevation Sensor Assembly (Installation Manual p.13)

ATA – Azimuth Time-of-Arrival

Auxiliary omni receiver aiding azimuth TOA measurements and multipath robustness.

Figure 2-8 ESA option without guy wires

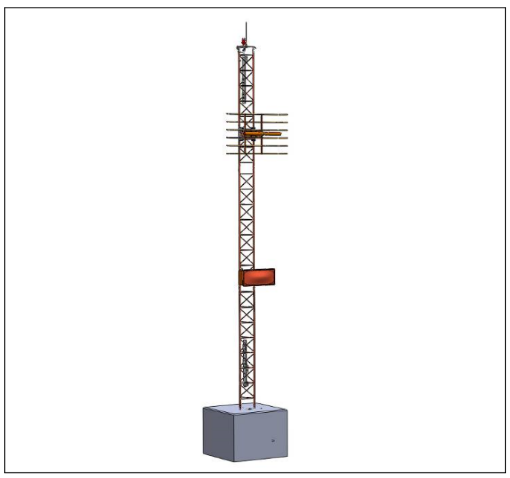

Figure 2-9 Uplink option without guy wires and without shelter attached

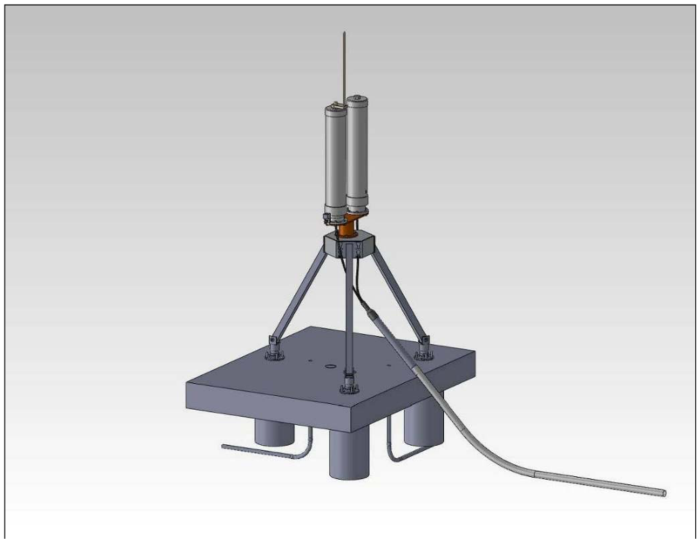

Figure 2-10 Interrogation with suppression option

NOTE: Interrogation must point towards the approach and ILS service volume; the suppression antenna beam must always point away from the approach direction.

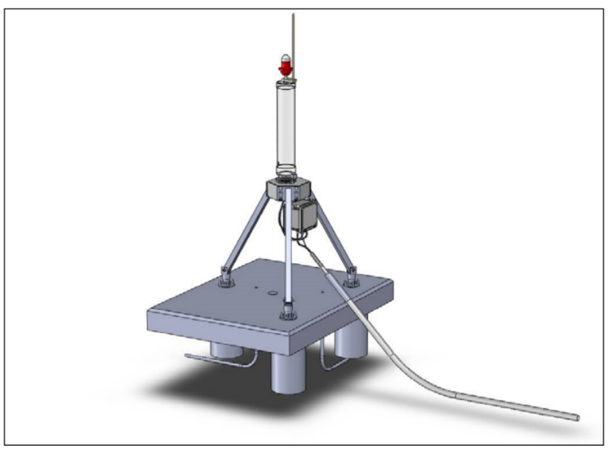

Figure 2-11 Cal/BIT with directional antenna, the antenna beam must point toward TLS

ATA – Azimuth Time-of-Arrival (Installation Manual p.15)

Cal/BIT – Calibration / Built-In-Test

1090 MHz test source used to validate sensor timing, phase, and sensitivity (BIT). Ensures integrity and simplifies return-to-service.

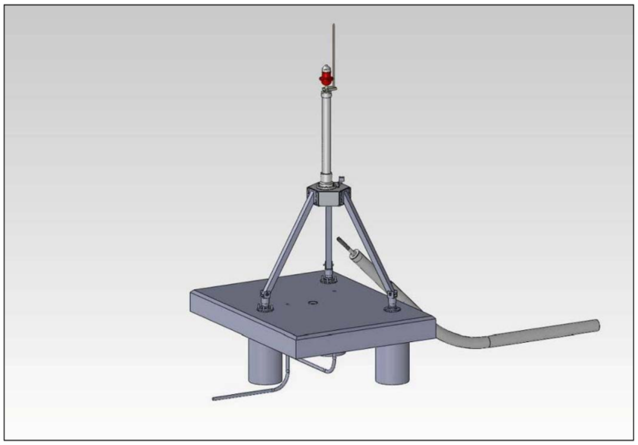

Figure 2-12 ATA with omni directional antenna

2.2.1 Equipment Locations

Due to the tower height, the ESA sensor location and its critical area should be considered first, followed by the ASA sensor and critical area, the ATA structure, the Cal/BIT structure, and then the Shelter. Final placement of components is the result of an iterative process that takes into consideration the placement requirements in Table 2-1. The critical area should also be considered early in the siting process in order to place the critical area in a location where it has minimal effect on operations. Consider the following important points when siting the TLS antennas:

- The TLS Shelter is the only point where utility power must be provided
- The critical area should be defined to have minimal impact on airport operations.
- The tall antennas of the ESA and Uplink should be outside of the Runway Safety Area and Obstacle Free Zone; other antennas are frangible and may be placed in these areas as allowed by ICAO and FAA because they are “fixed-by-function”
- For narrow properties the towers can be sited within the RSA and OFZ with waivers if no other options are available.
- Attempt to site the antennas to have minimal trenching, especially under runways; make use of existing conduits. Minimize trenching distance or complexity to bury the fiber-optic cables between the sensors and the TLS Shelter.

2.2.2 Shelter Placement

The Shelter location is typically between the ESA and ASA sensors. The shelter can be installed at any convenient location on or near the airport that satisfies the following conditions:

- The localizer, glide slope, and interrogation antennas must provide line of sight coverage to the required TLS service volume.
- Minimize trenching distance and complexity to bury the fiber-optic and power cables between the TLS shelter, the sensors and Cal/BIT.

Cal/BIT – Calibration / Built-In-Test (Installation Manual p.16)

2.2.3 TLS Component Siting

The TLS construction drawing package should be updated to show the TLS component siting. The drawing package should then be released to the contractor. The final as-built drawing package should be archived as a final document of the installation.

2.2.3.1 Concrete Foundations

All concrete foundations for TLS installation shall be completed in accordance with the site construction drawing package, unless modifications are needed to add stability in consideration of local soil conditions.

2.2.3.2 Power Requirements

The TLS electronics receive conditioned and protected AC power from the shelter UPS. Total power draw by option is listed in the figure below.

During the site design, conductor size is determined such that the voltage drop between the shelter and each component is less than 10% of the voltage.

Utility service varies according to options installed. Options include dual rack or single, split unit A/C configuration or standard.

Table 2-2 Electrical Requirements

Options	KVA		Total KVA per option	VOLTS	AMPS SERVICE	AMPS for TLS total rack(s) plus ECU's
Dual rack	Configuration	Power draw				
	UPS rack 1	2.9				
	UPS Rack 2	2.9				
	UPS Network	2.9	8.7	240 / 120	36.3 / 72.5	
Extreme hot	Split A/C	6.0	12.0	240 / NA	50 / NA	86.3 / 123
Standard A/C		0.54	0.54	NA / 120	NA / 4.5	40.8
Shelter heat	Standard	1.0	1.0	240 / 120	4.2 / 8..3	49
Deicing	Antennas	13.0	13.0	240 / NA	54 / NA	103
Single rack						
	UPS rack 1	2.9	2.9			
Extreme hot	Split A/C	6.0	12.0	240 / NA	50 / NA	52.9
Standard	a/c	0.54	0.54	NA / 120	4.5	7.4
Shelter heat	Standard	1.0	1.0	240 / 120	4.2 / 8..3	11.6
Deicing	Antennas	13.0	13.0	240 / NA	54 / NA	65.6

2.2.3.3 Installation Verification

For fixed TLS (not Transportable TLS) record the following verifications:

- System components are installed at locations per the construction drawing package.
- Concrete foundations have been constructed per the site construction drawing package.
- As-built redlines, where necessary, have been documented.

Cal/BIT – Calibration / Built-In-Test (Installation Manual p.17)

Interrogation & Suppression Antennas

1030 MHz interrogations trigger Mode A/C replies. Optional P2 suppression shapes the active service volume.

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	Minimum	Standard	Maximum
ASA w.r.t Touch Down (TD)	TD-100m	TD-50m	TD+75m
ASA to ESA separation	50 m (option 30 m)	100 m	No spec
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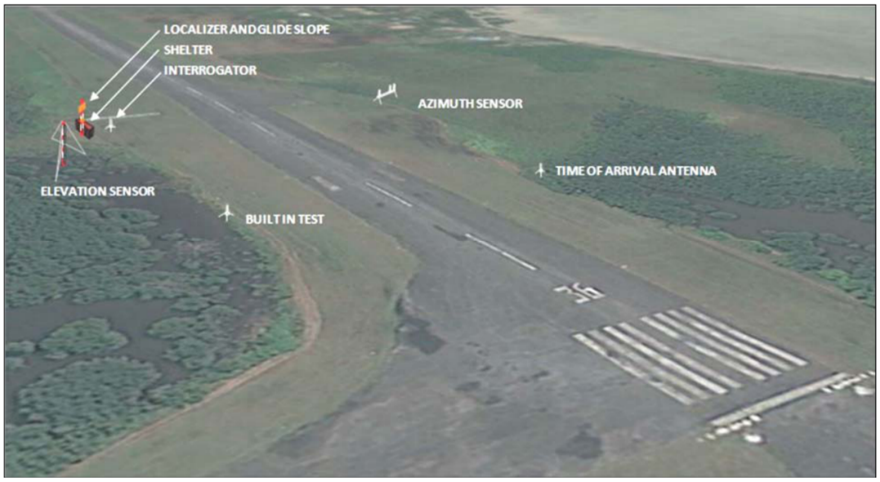
Interrogation & Suppression Antennas (Installation Manual p.11)



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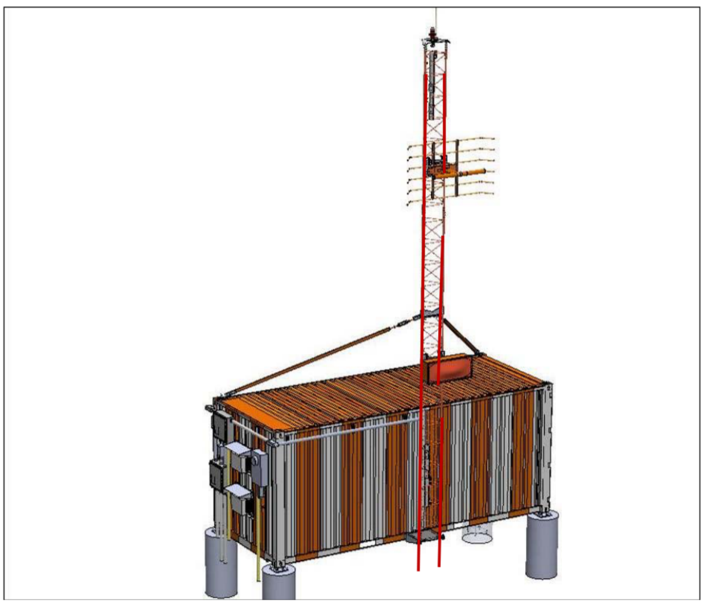
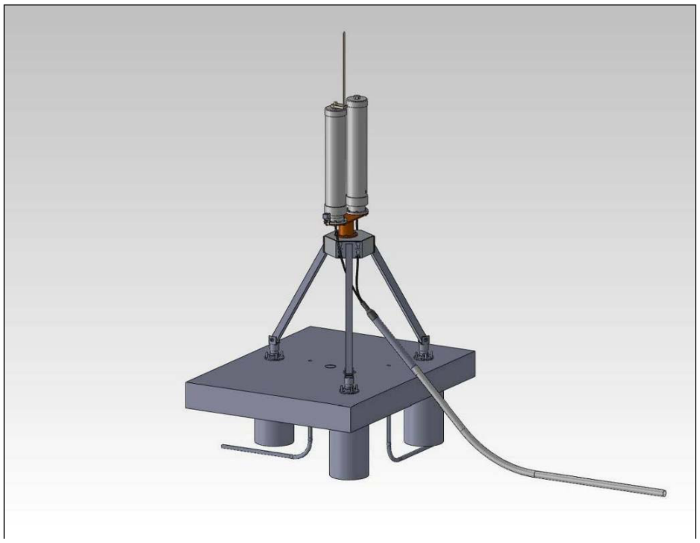


Figure 2-5 TLS shelter with hinged uplink antenna

Interrogation & Suppression Antennas (Installation Manual p.12)

Uplink – Localizer & Glide Slope

VHF localizer and UHF glide slope antennas broadcast ILS-like guidance tailored to the cleared aircraft.



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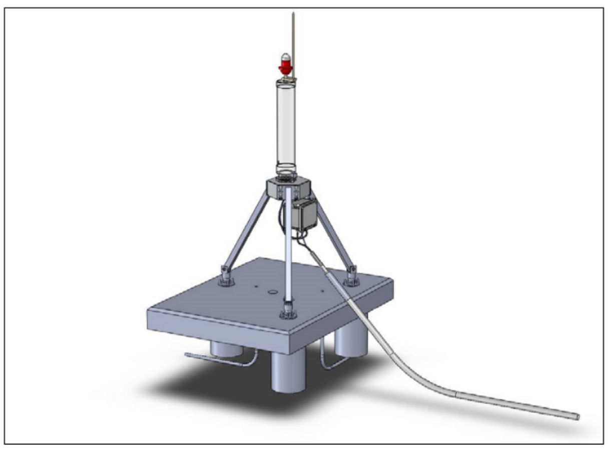
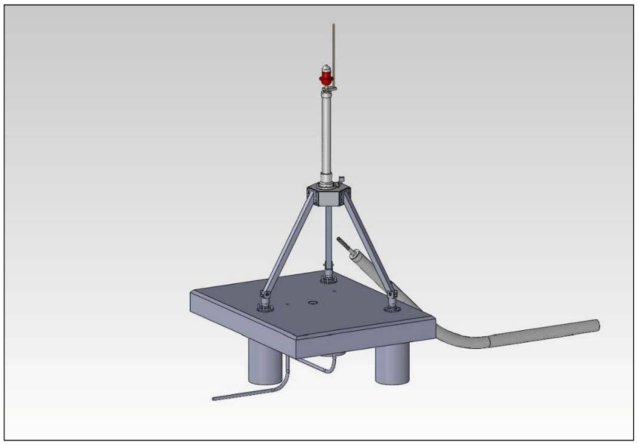


Figure 2-11 Cal/BIT with directional antenna, the antenna beam must point toward TLS



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Uplink – Localizer & Glide Slope (Installation Manual p.16)

System Relationships

• Interrogation stimulates transponder replies; ASA/ESA/ATA receive and estimate position. • Base station fuses measurements from sensors, runs integrity monitoring, and computes guidance. • Uplink transmits localizer and glide slope signals tuned per cleared aircraft. • Cal/BIT provides known test signals to validate sensors and timing.

Image Credits

Figures are extracted from the provided ANPC manuals (Installation and Operators) you shared, preserved for technical clarity.